

# Curriculum Vitae

Yue Ma — Experimental Nuclear Physicist

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## 1. Profile

Experimental nuclear physicist and Senior Research Scientist (permanent) at RIKEN, with more than 15 years of front-line research and leadership at major international facilities in Japan (J-PARC, SPring-8, KEK) and Germany (GSI, DESY), and a long record of training young researchers and building international collaborations. **Currently experiment spokesperson of J-PARC E73 (hypertriton lifetime), and proposal spokesperson of SPring-8 LEPS2 (nucleon short-range correlations) and the J-PARC antideuteron experiment.**

## 2. Academic Leadership & Appointments

- J-PARC E73, **hypertriton lifetime — Experiment spokesperson.** Beam time completed; the  ${}^4_{\Lambda}\text{H}$  lifetime and  ${}^3,4_{\Lambda}\text{H}$  production cross sections are published; the  ${}^3_{\Lambda}\text{H}$  lifetime analysis is ongoing.
- SPring-8 LEPS2, **nucleon short-range correlations — Proposal spokesperson.** Probing short-range correlations between nucleons with the  $\Lambda$  hyperon as a probe (ongoing).
- J-PARC antideuteron experiment — **Proposal spokesperson.** Antideuteron–nucleus interaction and annihilation mechanism (ongoing).

## 3. Signature Achievements

1. **Discovery of the  $K^-pp$  ( $\bar{K}NN$ ) kaonic nuclear bound state** — as a founding member, developed the multi-hit TDC and extracted the signal by polarization analysis, giving the first evidence for a deeply bound  $K^-pp$  state (J-PARC E15). Phys. Lett. B **789** (2019); Phys. Rev. C **102** (2020).
2. **Precise lifetime measurement of the  ${}^4_{\Lambda}\text{H}$  hypernucleus** — as spokesperson and corresponding author, proposed the key idea of tagging the fast  $\pi^0$  with a single high-energy  $\gamma$  and led the measurement,  $\tau = 206 \pm 8 \pm 12$  ps (J-PARC E73). Phys. Lett. B **845** (2023); **873** (2026).
3. **First observation of a  $\Lambda\pi$  structure near the  $\bar{K}N(I=1)$  threshold** — led the Belle data analysis as first author. Phys. Rev. Lett. **130** (2023).
4. **First application of TES microcalorimeters to hadronic-atom spectroscopy** — implemented the SDD data analysis and energy calibration at sub-eV precision (J-PARC E62). Prog. Theor. Exp. Phys. **2016**, 091D01; Phys. Rev. Lett. **128** (2022).

## 4. Education

|           |                                                       |
|-----------|-------------------------------------------------------|
| 2006–2009 | <b>Doctor of Science</b> , Tohoku University (Sendai) |
| 2004–2006 | M.S. in Physical Science, Tohoku University (Sendai)  |
| 1998–2002 | <b>B.S. in Nuclear Physics</b> , Lanzhou University   |

## 5. Appointments

|           |                                                             |
|-----------|-------------------------------------------------------------|
| 2012–now  | <b>Senior Research Scientist (permanent)</b> , RIKEN        |
| 2009–2012 | Postdoctoral Researcher, Helmholtz Institute Mainz, Germany |
| 2002–2004 | Research Trainee, Institute of Modern Physics, CAS          |

## 6. Grants and Fellowships

|           |                                                                                                                                                                                                               |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026–2027 | Grant-in-Aid for Transformative Research Areas (A), Publicly Offered Research: search for a $\Sigma$ -hypertriton ( $^3_\Sigma\text{H}$ ) bound state at J-PARC (within E. Hiyama's research area). <b>PI</b> |
| 2024–2028 | Grant-in-Aid for Scientific Research (A): $\rho$ -meson mass modification in nuclei with photon beams. Co-investigator                                                                                        |
| 2023–2025 | RIKEN Incentive Research Project: exploring matter–antimatter interactions. <b>PI</b>                                                                                                                         |
| 2017–2020 | JSPS Grant-in-Aid for Young Scientists (A) 17H04842: solving the hypertriton lifetime puzzle. <b>PI</b>                                                                                                       |

## 7. Student Supervision

|           |                                                        |
|-----------|--------------------------------------------------------|
| 2017–2020 | C. Han (PhD), Univ. of Chinese Academy of Sciences     |
| 2012–2014 | Q. Zhang (PhD), Lanzhou University                     |
| 2010–2011 | D. Lin (PhD); Christina, Pascal (M.S.), Univ. of Mainz |

## 8. Service & Invited Talks

|           |                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------|
| 2019–2021 | Member, RIKEN Scientists' Assembly Steering Committee; co-ordinator / chief coordinator, RIKEN Incentive Research Fund (Physics) |
| 2022      | Invited talk, 14th Int. Conf. on Hypernuclear & Strange Particle Physics (HYP 2022, Prague)                                      |
| 2025      | Invited talk, 15th HYP (HYP 2025, Tokyo)                                                                                         |

## 9. Teaching & Languages

|                  |                                                                                      |
|------------------|--------------------------------------------------------------------------------------|
| 2018/23/24       | Invited lecturer, Hadron Physics summer course, Univ. of Chinese Academy of Sciences |
| <b>Languages</b> | Chinese (native) · English (fluent) · Japanese (professional working proficiency)    |

## 10. Selected Publications (by leading role and impact; authorship indicated)

More than **100 publications** (including Belle / Belle II collaboration papers, see the final section); the following are first-author / corresponding-author / spokesperson or otherwise leading contributions.

Note: hadron-physics papers are usually listed alphabetically by surname; the **first author** / **corresponding author** / **spokesperson** tags below mark the actual leading role.

1. **Y. Ma et al.**, *First observation of  $\Lambda\pi^+$  and  $\Lambda\pi^-$  signals near the  $\bar{K}N(I=1)$  mass threshold in  $\Lambda_c^+ \rightarrow$*

- $\Lambda\pi^+\pi^+\pi^-$  decay, Phys. Rev. Lett. **130**, 151903 (2023) [Belle]. [\[First author\]](#)
2. T. Akaishi et al., *Precise lifetime measurement of  ${}^4_\Lambda\text{H}$  hypernucleus using a novel production method*, Phys. Lett. B **845**, 138128 (2023). [\[Spokesperson & corresponding author\]](#)
  3. T. Akaishi et al., *Measurement of  ${}^{3,4}\text{He}(K^-, \pi^0){}_\Lambda^3,4\text{H}$  reaction cross section and evaluation of hypertriton  $\Lambda$  binding energy*, Phys. Lett. B **873**, 140163 (2026). [\[Spokesperson\]](#)
  4. Y. Ma et al.,  *$\gamma$ -ray spectroscopy study of  ${}^{11}_\Lambda\text{B}$  and  ${}^{12}_\Lambda\text{C}$* , Eur. Phys. J. A **33**, 243 (2007). [\[First author\]](#)
  5. S. Ajimura et al.,  *$K^-pp$ , a  $\bar{K}$ -meson nuclear bound state, observed in  ${}^3\text{He}(K^-, \Lambda p)n$  reactions*, Phys. Lett. B **789**, 620 (2019). [\[Major author\]](#)
  6. T. Yamaga et al., *Observation of a  $\bar{K}NN$  bound state in the  ${}^3\text{He}(K^-, \Lambda p)n$  reaction*, Phys. Rev. C **102**, 044002 (2020). [\[Major contributor\]](#)
  7. T. Hashimoto et al., *Strong-interaction effects in kaonic-helium isotopes at sub-eV precision with X-ray microcalorimeters*, Phys. Rev. Lett. **128**, 112503 (2022). [\[Major contributor\]](#)
  8. B. S. Henderson et al., *Hard two-photon contribution to elastic lepton-proton scattering (OLYMPUS)*, Phys. Rev. Lett. **118**, 092501 (2017). [\[Major contributor; led the PbF<sub>2</sub> luminosity monitor\]](#)
  9. K. Hosomi et al., *Precise determination of  ${}^{12}_\Lambda\text{C}$  level structure by  $\gamma$ -ray spectroscopy*, Prog. Theor. Exp. Phys. **2015**, 081D01 (2015). [\[Second author\]](#)

## 11. Belle / Belle II Collaboration Papers

As a member of the Belle / Belle II collaboration since **2023**, co-author of **70+** collaboration papers; representative papers by theme:

**I.  $B$ -meson CP violation & CKM matrix elements** Time-dependent CP violation ( $\sin 2\phi_1$ , etc.),  $|V_{cb}|/|V_{ub}|$ , and the CKM angle  $\phi_3$ .

1. *Measurement of  $\sin 2\phi_1$  in  $B^0 \rightarrow J/\psi K_S^0$  decays with a graph-neural-network flavor tagger*, Phys. Rev. D **110**, 012001 (2024) [Belle II].
2. *Measurement of CP violation in  $B^0 \rightarrow K_S^0 \pi^0$  decays*, Phys. Rev. Lett. **131**, 111803 (2023) [Belle II].
3. *Determination of  $|V_{cb}|$  using  $\bar{B}^0 \rightarrow D^{*+} \ell^- \bar{\nu}_\ell$  decays*, Phys. Rev. D **108**, 092013 (2023) [Belle II].
4. *Measurement of inclusive  $B \rightarrow X_u \ell \nu$  partial branching fractions and  $|V_{ub}|$* , Phys. Rev. D **113**, 032004 (2026) [Belle II].
5. *Determination of the CKM angle  $\phi_3$  from a combination of Belle and Belle II results*, J. High Energy Phys. **10** (2024) 143 [Belle/Belle II].

**II. Rare decays & lepton-flavour universality** First evidence for  $B^+ \rightarrow K^+ \nu \bar{\nu}$ ,  $R(D^*)$  and other LFU tests, and  $\tau$  lepton-flavour-violation searches.

1. *Evidence for  $B^+ \rightarrow K^+ \nu \bar{\nu}$  decays*, Phys. Rev. D **109**, 112006 (2024) [Belle II].
2. *Test of light-lepton universality with a measurement of  $R(D^*)$  using hadronic  $B$  tagging*, Phys. Rev. D **110**, 072020 (2024) [Belle II].
3. *First measurement of  $R(X_{\tau/\ell})$  as an inclusive test of the  $b \rightarrow c\tau\nu$  anomaly*, Phys. Rev. Lett. **132**, 211804 (2024) [Belle II].
4. *Tests of light-lepton universality in angular asymmetries of  $B^0 \rightarrow D^{*-} \ell \nu$  decays*, Phys. Rev. Lett. **131**, 181801 (2023) [Belle II].
5. *Measurement of the  $B^+ \rightarrow \tau^+ \nu_\tau$  branching fraction with a hadronic tagging method*, Phys. Rev. D **112**, 072002 (2025) [Belle II].
6. *First search for  $B \rightarrow X_s \nu \bar{\nu}$  decays*, Phys. Rev. Lett. **136**, 231801 (2026) [Belle II].
7. *Search for lepton-flavor-violating  $\tau$  decays to  $\ell\alpha$* , J. High Energy Phys. **08** (2025) 155 [Belle].
8. *Search for a  $\tau^+\tau^-$  resonance in  $e^+e^- \rightarrow \mu^+\mu^-\tau^+\tau^-$  events*, Phys. Rev. Lett. **131**, 121802 (2023) [Belle II].

**III. Charm/strange baryon spectroscopy & exotic hadrons**(closely aligned with my own research)  $\Xi_c$  decays and CP, the first observation of  $D_{s0}^*(2317)$  radiative decay,  $P_{cs}$  pentaquarks, and  $\Xi^0 p/\Omega^- p/\Omega^- n$

dibaryon searches.

1. *Observation of the radiative decay  $D_{s0}^*(2317)^+ \rightarrow D_s^{*+}\gamma$* , Phys. Rev. Lett. **136**, 241901 (2026) [Belle/Belle II].
2. *Search for  $P_{cs}(4459)$  and  $P_{cs}(4338)$  in  $\Upsilon(1S, 2S)$  inclusive decays*, Phys. Rev. Lett. **135**, 041901 (2025) [Belle/Belle II].
3. *Search for CP violation in  $\Xi_c^+ \rightarrow \Sigma^+ h^+ h^-$  and  $\Lambda_c^+ \rightarrow p h^+ h^-$  decays*, Phys. Rev. D **113**, 032017 (2026) [Belle II].
4. *Observation of the singly Cabibbo-suppressed decays  $\Xi_c^+ \rightarrow p K_S^0, \Lambda \pi^+, \Sigma^0 \pi^+$* , J. High Energy Phys. **03** (2025) 061 [Belle/Belle II].
5. *Measurement of branching fractions of  $\Xi_c^0 \rightarrow \Xi^0 \pi^0, \Xi^0 \eta, \Xi^0 \eta'$* , J. High Energy Phys. **10** (2024) 045 [Belle/Belle II].
6. *Observation of  $B^+ \rightarrow \Sigma_c(2455)^{++} \bar{\Xi}_c^-$  and  $B^0 \rightarrow \Sigma_c(2455)^0 \bar{\Xi}_c^0$* , Phys. Rev. D **112**, L051101 (2025) [Belle/Belle II].